A Review on Social Media Threats and Its Emerging Solutions

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***Abstract*—The rapid growth of social media platforms has completely changed how we communicate around the world. However, it has also led to many cyber threats. This paper looks at the rising cyber threats on social media, including cyberbullying, deepfakes, and illegal drug trafficking. These issues exploit user weaknesses and spread harm globally. It includes a survey of recent studies and examines effective detection methods like deep learning models, multimodal analysis, and AI-driven systems to reduce these risks. It also discusses key challenges such as real-time processing, support for multiple languages, and ethical concerns. Emerging trends like explainable AI and crowdsourced data are noted as well. The paper suggests future paths for strong, flexible solutions to create safer digital spaces. Ultimately, it emphasizes the need for collaboration across different fields to tackle the changing online threats.**

***Index Terms*—social media threats, cyberbullying detection, deepfake identification, illicit drug trafficking, AI-based solutions, cybersecurity challenges**

I. INTRODUCTION

With rapid technological growth, cyberattacks have become a serious global concern, especially through social media platforms like Facebook, Instagram, TikTok, and Twitter. Their vast reach has enabled attackers to exploit users, spread disinformation, and conduct illicit activities. The growth of social media and cyberattacks are closely linked, demanding stronger security, policy frameworks, and user awareness to create safer digital spaces.As shown in , and you will see that the rise in cyberattacks goes hand in hand with the increasing popularity of social media platforms. Among the various attacks identified, illicit drug activities, cyberbullying, and deepfakes are the most commonly observed threats, yet when combined, they remain relatively underexplored in existing research.

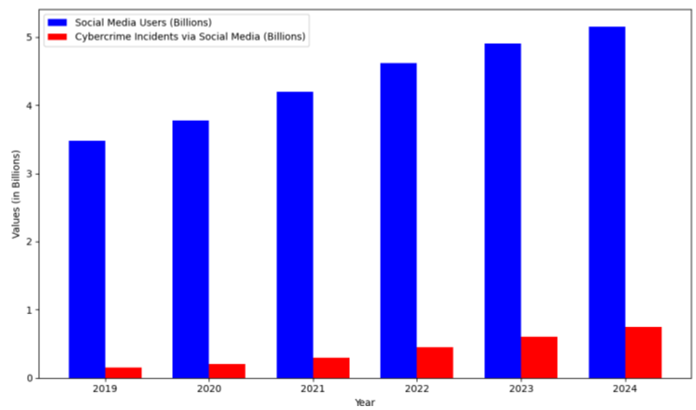

Fig. 1. Growth of Social Media Users vs. Cybercrime Incidents via Social Media

II. LITERATURE SURVEY

A. Cyber Bullying

S.M. Fati, Amgad Muneer, and Ayed Alwadain [1] proposed a deep learning framework for detecting cyberbullying on Twitter. Their approach combines an attention mechanism with Continuous Bag of Words (CBOW) feature extraction. The model captures contextual meaning well and shows better accuracy than traditional machine learning methods. The attention mechanism also makes it easier to understand by highlighting key features that affect predictions. However, the model relies on English-language datasets, which limits its use in different languages and cultures. Additionally, CBOW embeddings might not fully capture changes in online language or implicit aggression. To tackle these issues, incorporating multilingual datasets, cross-lingual or transformer-based embeddings, and flexible contextual modeling could greatly

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TABLE I
SUMMARY OF CYBERBULLYING STUDIES

Literature	Methods Used	Results	Research Gap
MDPI,2023 [1]	GCN, GAT, text normalization, TF-IDF, and adjacency matrix for graph-based rumor detection.	94.87% accuracy, GAT: 87.79%, outperforming CNN	Limited GraphSAGE/heterogeneous GNN use; no multilingual testing
IEEE, 2025 [2]	BiLSTM + BERT (Multilingual-Cased); Text normalization (Hungarian data)	Effective for Hungarian cyberbullying detection	No testing on code-mixed/dialect languages; No real-time deployment analysis
IEEE, 2025 [3]	GCN, GAT; Text normalization, TF-IDF, adjacency matrix	GCN: 94.87% accuracy, GAT: 87.79%; Outperformed CNN	Limited GraphSAGE/heterogeneous GNNs; No multilingual testing
IEEE, 2025 [4]	K-Means clustering + NLP emotion detection (fear, anger, anxiety)	Spikes in negative emotion predicted cyberbullying; Early warning potential	No integration with geolocation/demographics; No real-time capabilities

enhance the strength, inclusivity, and practical use of these detection systems.

B. To´th, S.A. Laczi, and V. Po´ser [2] treated cyberbullying detection as a multi-class classification problem. They combinned a fine-tuned BERT-Base-Multilingual-Cased model with a Bidirectional LSTM (BiLSTM) network. The multilingual BERT component produces contextualized embeddings that capture meaning and language differences. The BiLSTM layer models sequential dependencies to improve context flow and understand the timing of online discussions. Preprocessing steps like tokenization, normalization, and noise removal en- sure data consistency before model training. The design of the framework allows effective detection across various types of bullying, going beyond binary detection systems in detail and clarity. However, even with strong cross-language capabilities, the model's focus on text-based cues only may restrict its flexibility to handle the mixed media of today's social media threats. Abusive or harmful intent can be found in images, videos, or synthetic content like deepfakes. Tackling this limitation through multimodal fusion and incorporating visual-linguistic representations could strengthen these systems. This approach would expand their use in detecting digital threats beyond just textual cyberbullying.

M. Ragab, A. Muneer, and A. Alwadain [3] created a stacking ensemble framework that includes a boosted BERT model for detecting cyberbullying on Twitter and Facebook. This system combines several learning algorithms to take advantage of their strengths, achieving impressive results: 97.4% accuracy, 95.0% precision, and a 96.4% F1-score. It significantly outperforms traditional single-model approaches. The ensemble design makes use of BERT's ability to understand context, while boosting techniques improve its predictive stability and generalization across different datasets. This combined setup shows how effective ensemble deep learning can be in recognizing the subtle language and meaning behind online harassment. However, the model mainly works with text, which limits its ability to adapt to the changing nature of multimodal social media threats where harmful behavior might appear in memes, videos, or deepfake content. Expanding these ensemble strategies to include multimodal and cross-platform data could strengthen resilience and adaptability. This would enable a more thorough defense against new digital threats like cyberbullying, misinformation, and the spread of harmful content.

P. Privietha and colleagues [4] proposed a multi-stage framework for detecting cyberbullying that combines transformer-based embeddings with sequential neural networks. The process starts with multilingual data preprocessing, including tokenization, normalization, and noise reduction. Then, it extracts contextual embeddings using transformer models like multilingual BERT. These embeddings move through sequence modeling layers such as BiLSTM, GRU, or attention mechanisms to capture temporal and semantic links in online discussions. The model trains as a multi-class classifier, distinguishing between different types of bullying behavior with suitable loss functions and class balancing strategies. While this setup shows strong cross-lingual adaptability and better detection accuracy across various bullying types, its focus on textual cues alone restricts its ability to fully understand the complex and changing nature of social media threats. These threats often involve implicit aggression, coded language, and rapidly evolving communication patterns. Improving future cyberbullying detection systems in dynamic online settings could involve addressing these limitations through adaptive linguistic modeling and context-aware analysis.

Table I summarizes the key studies on cyberbullying detection discussed above. It provides a comparative overview of the methods employed, their results, and identified research gaps, serving as a concise reference to understand the current techniques and challenges in detecting and mitigating cyberbullying on social media platforms.

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B. Deepfake

A deepfake is a digitally manipulated video or image that convincingly replaces a person's likeness or actions using advanced AI algorithms. The rise of highly realistic video deepfakes poses significant threats to privacy, security, and trust in media. These manipulated videos can misrepresent individuals and spread misinformation, with legal, ethical as well as social consequences [. As deepfakes become more accessible, effective detection methods are critical to mitigate their malicious use.

G. Petmezas and colleagues [5] introduced a deepfake detection framework that combines Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Transformer architectures to improve identity verification through facial feature analysis. Unlike standard artifact-based methods, their approach relies solely on real video samples from the VoxCeleb2 dataset. This allows the model to learn genuine facial dynamics and detect inconsistencies when tested against manipulated content. This strategy boosts generalization across different deepfake generation techniques and performs well on key datasets like FaceForensics++, Celeb-DF, and DFD. The hybrid design effectively uses CNNs for spatial feature extraction, LSTMs for modeling temporal dependencies, and Transformers for learning contextual representations. However, the framework shows less reliability in cross-platform settings and real-time social media environments, where changes in compression, resolution, and evolving manipulation strategies create significant challenges. These issues emphasize the ongoing need for flexible and scalable deepfake detection methods that can keep up with the rapidly changing nature of synthetic media spread on digital platforms.

H. Alazzam and colleagues [6] proposed a hybrid deep learning framework that combines Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). They optimized this framework through Particle Swarm Optimization (PSO) to improve deepfake detection accuracy. The methodology includes thorough preprocessing of video frames, which involves segmentation, resizing, normalization, stabilization, and audio extraction. This is followed by extracting spatial features using CNNs and modeling temporal sequences with RNNs. PSO acts as an optimization strategy to fine-tune network settings, such as the number of layers, neuron counts, and activation functions. It does this through ongoing feedback on the model's performance. The proposed system achieved strong results across standard metrics like accuracy, precision, and recall. It demonstrated a reliable ability to distinguish between genuine and manipulated content by using both spatial and temporal information. However, even with its success in controlled experimental settings, the model's ability to adapt to real-world social media data is limited. This limitation arises from variations in compression formats, user-generated noise, and the increasing sophistication of deepfake generation techniques. This highlights the need for more resilient and context-aware detection methods to tackle the changing nature of synthetic media on online platforms.

S. Mumtaz and colleagues [7] introduced a patch-wise deep learning framework for detecting deepfake facial manipulations made by different AI systems. The model works by analyzing localized image patches instead of entire frames. This approach helps it spot subtle artifacts and inconsistencies that global methods often miss. Evaluated on benchmark datasets like Celeb-DF and DFDC, the method achieved high classification accuracy by capturing fine details across generative model outputs. The study also includes attention-based mechanisms and data augmentation techniques to improve the model's focus on key facial areas and strengthen its robustness against changes in synthetic content. However, the authors note that while analyzing patches improves sensitivity to local distortions, the model has limited generalization across various deepfake creation methods and real-world social media situations. In these cases, compression, filters, and post-processing often hide important forensic clues. This limitation highlights the need for detection systems that can reliably handle the growing complexity of AI-generated media threats.

By examining language patterns, S. Sadiq and colleagues [8] used a deep learning framework along with FastText embeddings to tell apart machine-generated tweets from genuine ones. Their method takes advantage of FastText's ability to capture subword details. This lets the model spot style irregularities often found in synthetic text. When tested on Twitter datasets with both human-written and algorithmically created tweets, the model showed good accuracy in finding automated deepfake content. However, the study observed a decline in performance when faced with advanced language models that can imitate human meaning and conversation flow. This suggests that static embedding methods might struggle with rapidly changing generative algorithms. Additionally, the research focuses only on text, ignoring multimodal deepfake threats that involve altered audio or visual content alongside text distribution. Given the growing issues of cyberbullying, misinformation campaigns, and illegal activities supported by cross-modal media on social platforms, adding multimodal analysis and flexible embedding strategies could greatly improve the effectiveness and real-world use of deepfake detection systems.

J. Choi and colleagues [9] introduced an identity-centric audio deepfake detection framework that uses speaker-specific voice characteristics to improve cross-dataset generalization, especially in benchmarks like ASV Spoof. Instead of depending only on global acoustic patterns, their method focuses on personalized vocal traits, including timbre consistency, pitch modulation patterns, and spectral envelope signatures. These elements help distinguish real speech from synthesized audio. Experimental results showed that identity-aware modeling significantly boosts performance against various deepfake generation techniques. This underscores the importance of maintaining speaker identity cues in detection processes. However, the authors also noted that the system becomes less effective when faced with noisy, compressed, or low-quality recordings, which can distort identity features. Moreover, performance differences across diverse datasets reveal

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challenges in keeping robustness in real-world social media scenarios, where audio uploads differ widely in format and quality. To improve resilience against new audio-related cyber threats, such as impersonation scams, manipulated voice-based harassment, and coordinated misinformation, they suggest integrating self-supervised representation learning, additional acoustic descriptors, and multimodal signals. This combination would help create a more flexible and context-aware detection framework.

G. Sivaraman and his team [10] looked at how voiced and unvoiced speech segments can reveal small acoustic inconsistencies in synthetic audio. They studied changes in time and frequency in datasets like WaveFake and ASVSpoof. The research showed that deepfake speech often has irregular shifts between voiced and unvoiced parts, which are different from natural human speech patterns. Their findings underscore the importance of detailed acoustic feature analysis in improving audio deepfake detection accuracy. However, the authors pointed out some limitations. They noted challenges in defending against deepfakes that are specifically designed to imitate natural speech variations, which weakens the system's effectiveness in uncontrolled settings. Additionally, their framework focuses only on audio and ignores the connected nature of manipulation efforts on social media, where voice, text, and visual content often support each other. To develop more flexible detection systems that work well against large-scale harmful activities online, such as impersonation scams, coordinated harassment, and illegal network promotion, they suggest combining spectral analysis with multimodal cues like linguistic context and visual metadata. This approach could create a more integrated detection system [10].

The following Table II summarizes the key studies on deepfake detection discussed above, along with additional relevant research.

C. *Illict Drugs Dealing*

Illicit drug trafficking involves the illegal sale and distribution of banned substances. This activity poses serious threats to public health and social stability. It is often connected to crime, violence, and corruption. With the rise of social media, drug dealers use online platforms to anonymously advertise and coordinate transactions. This makes detection more difficult.

C. Hu and colleagues [12] created a framework to detect drug trafficking events on Instagram. This framework combines textual captions, image content, and related hashtags. Their model uses CNNs to extract visual features and RNN-based encoders to identify linguistic patterns. It also includes embedding representations for the meanings of hashtags, allowing for multilabel classification of trafficking-related activities like promotion, sale, and usage. This fusion of different types of information showed strong detection performance in real social media conditions. However, the system struggles with scalability due to the high computational costs of processing various data streams, which limits real-time use. Additionally, the model relies heavily on cues visible on the

TABLE II
SUMMARY OF DEEPFAKE DETECTION STUDIES

Literature	Methods Used	Results	Research Gap
Springer, 2025 [5]	CNN, LSTM, Transformer	High accuracy on FaceForensics++	Struggles with cross-platform variations; needs more generalized models
MDPI, 2024 [6]	CNN-RNN with PSO	Promising results on DeepfakeTIMIT, UADFV	Challenges in real-time detection and dataset diversity; suggests multimodal analysis
MDPI, 2024 [7]	Patch-wise deep learning, Vision Transformer (ViT), Baseline: ResNet-50, VGG-16	High accuracy on Celeb-DF, DFDC; 99.90% accuracy for ViT	Poor generalization across deepfake algorithms; suggests hybrid architectures
IEEE, 2023 [8]	DT, LR, AC, SGC, RF, GBM, ETC, NB, LSTM, CNN-LSTM; CNN with FastText embeddings	93% accuracy of CNN with FastText; outperformed baseline models	Difficulties with sophisticated text generators; suggests updates to embeddings and multimodal analysis
IEEE, 2025 [9]	Contrastive learning with voice identity features	Robust performance against unseen models on ASVSpoof	Focuses on audio deepfakes only; limited to voice identity features; suggests multimodal data
IEEE, 2025 [10]	Graph attention-based AASIST, signal periodicity analysis	6.62% EER (unvoiced), 5.82% EER (combined) on WaveFake, ASVSpoof	Gaps in detecting adversarial attacks; suggests combining spectral analysis with other modalities
IEEE, 2025 [11]	ResNet50, LSTM for spatial-temporal modeling	High accuracy on FaceForensics++, DFDC	Needs scalable solutions for large-scale data; suggests domain-specific features and multimodal analysis

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platform. However, illicit networks often switch to coded language or hidden communication methods. The authors suggest adding auxiliary data sources, such as transactional indicators or geolocation traces, to improve monitoring and extend the model's use in changing illicit online markets.

E. Ferrara [13] studied geolocated social media posts to track drug cartel activity by identifying patterns in where and when posts occur. The study uses spatial clustering algorithms to find areas with high levels of drug-related discussions. It also applies time-series predictive models to estimate the chances of new cartel hotspots appearing. By mapping geolocation data to actual regions, the approach turns online activity into useful information for law enforcement and policy measures. However, the system faces challenges due to incomplete or incorrect location data. Many posts either do not have geotags or use misleading user-entered locations. To improve reliability and expand coverage, the author suggests adding other sources like satellite imagery and financial transaction records to create a stronger monitoring system.

Fisher A and colleagues [14] introduced a machine learning framework to automatically detect drug-related harms like addiction, overdose, and illegal distribution from social media posts. Their method uses NLP-based named entity recognition to spot mentions of substances, topic modeling to classify discussion themes, and sentiment analysis to understand risk-related language. This allows the system to identify harmful behaviors and emerging patterns in drug use, showing strong results with structured datasets. However, the model has trouble with contextual nuances such as sarcasm, slang, and coded language. It also only analyzes text, while real-world drug-related content often appears in images or videos. To enhance its effectiveness across various online situations, the authors suggest expanding the approach to include visual and video-based cues along with linguistic signals. This would help better capture hidden or changing signs of drug-related harm.

Table III summarizes the key studies on illicit drug dealing detection discussed above, along. It provides a comparative overview of the methods used,ir results, and identified research gaps, serving as a concise reference to understand techniques and challenges in detecting and mitigating illicit drug activities on social media platforms.

III. Challenges and Future Trends

Key shared challenges identified includes real-time detection, multilingual scalability, handling nuanced content, and addressing ethical concerns. Real-time detection remains a significant hurdle, as the survey highlights scalability and processing bottlenecks in multimodal models.By facilitating dynamic model updates and effective threshold modification, the suggested application of reinforcement learning and incremental learning reduces these problems and improves real-time responsiveness. Another recurring problem is multilingual scalability, which hinders models' capacity to generalize because there aren't enough high-quality, annotated datasets in a variety of languages [1]. To address this, the proposed approaches incorporate XLM-R, mBERT, and cross-lingual embeddings to improve cross-language transferability. Furthermore, detecting nuanced or obfuscated content—such as sarcasm, slang, or encrypted messages—continues to challenge traditional models. Advanced techniques like hierarchical attention mechanisms, sentiment-aware transformers, and adversarial training with GANs have been proposed to better capture these subtle and context-dependent cues. Lastly, ethical concerns such as user privacy, fairness, and the risk of false positives remain underexplored. Although the proposed models integrate interpretable components like attention mechanisms, the survey calls for further development of privacy-preserving frameworks such as federated learning and the broader adoption of explainable AI (XAI) to ensure ethical and responsible deployment of detection systems.

Emerging trends in social media-based cyber threat detection include multimodal integration, real-time systems, explainable AI, and crowdsourced data collection. Multimodal integration—combining text, images, audio, and metadata—has been shown to significantly enhance model robustness and contextual understanding across threat domains. The integration of IoT and real-time systems, including APIs and live threat intelligence feeds, is gaining traction, particularly in use cases such as phishing and data breach detection, where dynamic responsiveness is critical. Additionally, the adoption of Explainable AI (XAI) techniques, including attention mechanisms and fuzzy logic, contributes to improved interpretability and aligns with the growing demand for transparency in automated decision-making systems. Lastly, crowdsourced data collection, which has proven effective in agricultural monitoring, is emerging as a promising approach in cybersecurity. Engaging users to report cyberbullying, misinformation, or suspicious activity can provide valuable real-world data, facilitating early detection and mitigation of cyber threats.

IV. Conclusion

This survey research explores the changing landscape of cyberthreats on social media, including deepfakes, cyberbullying, and illegal drug trafficking. By examining 16 recent studies, it points out key research gaps, categorizes threats, and looks at different detection methods.

Techniques like CNN-LSTM-Transformers, Graph Neural Networks (GNNs), and hybrid approaches show strong performance in threat detection. However, they have difficulty with real-time processing and managing subtle content changes. The increasing complexity of threats highlights the need for flexible and scalable solutions.

The study suggests future research directions, such as multilingual detection models, geolocation-based analysis, real-time data integration, and ethical guidelines. By encouraging collaboration across different sectors, this research helps build strong and secure digital environments.

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TABLE III
SUMMARY OF ILLICIT DRUG DEALING DETECTION STUDIES

Literature	Methods Used	Results	Research Gap
IJETS, 2025 [15]	CNN, YOLO, OCR, hybrid ML for text + image detection	Real-time, automated image-based detection; reduces manual oversight	Focuses mainly on image processing; limited integration of textual social media cues
IEEE, 2025 [16]	Quantum Bi-LSTM with attention mechanism, variational quantum circuits (VQC)	Improved ADR detection on SMM4H dataset; better generalization and reduced computational needs	Application limited to ADR detection; not designed for illicit drug trafficking detection
IM, 2025 [12]	Knowledge-informed prompts using ChatGPT, Monte Carlo dropout for prompt optimization	12% improvement over baselines; effective against deceptive language	Relies on textual data only; may lack multimodal robustness
JMIR, 2023 [14]	Zero-shot learning on Twitter data; rule-based keyword filtering	84.5% specificity and 94.1% sensitivity vs expert labels	Limited to English/French tweets; lacks advanced NLP or multimodal fusion
GeoJournal, 2021 [13]	Temporal and spatial analysis of entity clusters from geolocated tweets	Enabled detection of cartel activity patterns and hotspots	Does not use advanced ML models; limited by reliance on manual clustering

TABLE IV
SUMMARY OF OPEN RESEARCH CHALLENGES ACROSS DOMAINS

Domain	Key Challenges	Open Research Focus Areas
Deepfake Detection	Generalization issues across platforms; weak audio deepfake handling; high computational cost	Lightweight multimodal systems; GAN-aware generalization; audio-visual integration for robustness
Illicit Drug Dealing	Overreliance on image features; geolocation metadata inconsistency; limited encrypted platform support	Integration of emoji-coded language; satellite or transaction-based location enrichment; cross-platform adaptability
Cyberbullying Detection	Lack of support for low-resource languages; under-modeled emotion dynamics; missing long-term context	Emotion-aware multilingual detection; demographic-informed response modeling; longitudinal risk tracking

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